THE SHEAF OF MEROMORPHIC FUNCTIONS ON A REDUCED SCHEME WITH FINITELY MANY COMPONENTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file divisors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Carrier window 3: statement and proof both lie inside the window 6000-8000 granted to this lane by the owner lane lane-s13 on 2026-10-14.

Editorial note. The proof is one sentence of specialisation with its reason stated; this card is the general input used by the meromorphic-section cards of this pool, where it enters as a navigation pointer.

Editorial note. Preamble: sources/stacks-divisors/retained-preamble.tex (88 lines, verbatim present).

Allowed standard prerequisites (editorial).

- The description of the sheaf of meromorphic functions through weakly associated points. Lemma lemma-meromorphic-weakass-finite, used at proof lines 6458-6461. Is the general statement that is specialised.
- On a reduced scheme the weakly associated points are the generic points. Lemma lemma-weakass-reduced, used at proof lines 6461-6463. Supplies the reason the specialisation applies.

Source reference. Original lemma 0EMF: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 18305–18305. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 0EME: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 18304–18304. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original section 01J5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2002–2002. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let X be a reduced scheme such that any quasi-compact open has a finite number of irreducible components. Let X^0 be the set of generic points of irreducible components of X. Then we have*

$$\mathcal{K}_X = \bigoplus\nolimits_{\eta \in X^0} j_{\eta,*}\kappa(\eta) = \prod\nolimits_{\eta \in X^0} j_{\eta,*}\kappa(\eta)$$

where $j_\eta : \mathrm{Spec}(\kappa(\eta)) \to X$ is the canonical map of Schemes, Section 01J5. Moreover

(1) *$\mathcal{K}_X$ is a quasi-coherent sheaf of $\mathcal{O}_X$-algebras,*

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(2) *for every quasi-coherent $\mathcal{O}_X$-module $\mathcal{F}$ the sheaf*
$$\mathcal{K}_X(\mathcal{F}) = \bigoplus\nolimits_{\eta \in X^0} j_{\eta,*}\mathcal{F}_\eta = \prod\nolimits_{\eta \in X^0} j_{\eta,*}\mathcal{F}_\eta$$
of meromorphic sections of $\mathcal{F}$ is quasi-coherent,
(3) *$\mathcal{S}_x \subset \mathcal{O}_{X,x}$ is the set of nonzerodivisors for any $x \in X$,*
(4) *$\mathcal{K}_{X,x}$ is the total quotient ring of $\mathcal{O}_{X,x}$ for any $x \in X$,*
(5) *$\mathcal{K}_X(U)$ equals the total quotient ring of $\mathcal{O}_X(U)$ for any affine open $U \subset X$,*
(6) *the ring of rational functions of X is the ring of meromorphic functions on X, in a formula: $R(X) = \Gamma(X, \mathcal{K}_X)$.*

Proof. This lemma is a special case of Lemma 0EMF because on a reduced scheme the weakly associated points are the generic points by Lemma 0EME. □